

All My Work

Your saved learning record for Genetic Diversity in Populations.

1 completed activity run. Required checks and formative practice are recorded separately.

Unfinished work: 0 required check(s), 1 practice run(s). Resume them from their original sections; they are not counted as completed.

Writing and flash-card self-assessments are not automatically graded. Completion is not a mastery certificate.

Mixed practice

9/18/2026, 12:21:45 AM · Elapsed 0m 1s · Formative practice

Review submitted responses and attempts

First-try result: 6/6. Later corrections are recorded separately.

Which term matches this description? Mating that is independent of the genotype being considered.

Options shown: random mating · Hardy–Weinberg equilibrium · $2pq$ · p^2

Attempt 1: random mating — Correct.

Reference: random mating

Correct this incorrect statement: A single attractive photograph proves a population is genetically secure.

Self-report: Study again.

Reference: Observations used to evaluate a population's condition and response to an action. Several measures are needed to distinguish mechanisms and outcomes.

Name the term described: The proportion of individuals with a specified genotype.

Attempt 1: genotype frequency — Correct.

Reference: genotype frequency

What does letter B identify?

Options shown: Observed aa fraction · Homozygous dominant frequency · Recessive allele frequency · Expected carrier count · Heterozygote frequency · Other allele frequency

Attempt 1: Recessive allele frequency — Correct.

Reference: Recessive allele frequency

Select the TWO accurate statements about populations, gene pools and genetic variation.

Options shown: A gene pool is the set of organs inside one organism. · A population boundary must be stated before counting frequencies. · Not all phenotypic variation is genetic. · Any increase in population size is automatically an allele-frequency change.

Attempt 1: true-1 → true-2 — Correct.

Reference: A population boundary must be stated before counting frequencies. · Not all phenotypic variation is genetic.

Put these steps in order: Calculating expected genotype frequencies

Options shown: Determine p and q · State the equilibrium and phenotype assumptions · Identify the observed or supplied quantity · Convert frequencies to counts if requested · Calculate expected genotype frequencies · Check totals and interpret the result

Attempt 1: Identify the observed or supplied quantity → State the equilibrium and phenotype assumptions → Determine p and q → Calculate expected genotype frequencies → Convert frequencies to counts if requested → Check totals and interpret the result — Correct.

Reference: Identify the observed or supplied quantity → State the equilibrium and phenotype assumptions → Determine p and q → Calculate expected genotype frequencies → Convert frequencies to counts if requested → Check totals and interpret the result

Which term fits this definition? A count actually supplied or measured in the sample.

Self-report: Study again.

Reference: observed count

Explain p^2 .

Self-report: Study again.

Reference: The expected frequency of one homozygous genotype under a two-allele Hardy-Weinberg model.

Identify the term illustrated: A small surviving subset carries only part of the earlier variation.

Attempt 1: bottleneck effect — Correct.

Reference: bottleneck effect

Only the proportion of recessive-phenotype individuals is known, with no equilibrium statement. What is missing before inferring q by a square root?

Self-report: Study again.

Reference: Justification for the genotype-frequency model. The observed recessive fraction is an aa frequency, but equating it with q^2 needs the model.

Frayer work

population — Draft

Meaning: An edited explanation.

Key features: A specific learner explanation 2.

Example: A specific learner explanation 3.

Non-example / common confusion: A specific learner explanation 4.

gene pool — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

genetic variation — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

microevolution — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

genotype frequency — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

allele frequency — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

heterozygote contribution — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

frequency sum — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

Collected optional responses

Biotechnology and gene pools

The comparison alone cannot isolate the proposed chemical effect. Hormone production and receptor response are distinct steps.

Textbook question work

Draft and saved question responses. Optional and ungraded.

No textbook question work yet.

Supported and independent practice